Day 12 Report: LLM Engineering

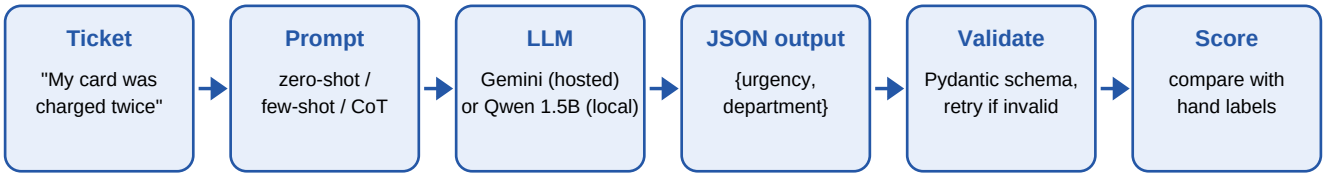
Prompting strategies, structured JSON output with validation, hosted vs local model

1. Goal

Learn to control LLMs reliably. Task: classify **50 customer support tickets** by **urgency** (high / medium / low) and **department** (billing, technical, account, shipping, general), and compare a hosted model with a small local model and three prompting styles.

2. How it works

Pipeline



Prompt style	What it is
Zero-shot	Instructions + the ticket, no examples
Few-shot	Same instructions + 5 labelled examples
Chain-of-thought (CoT)	Model writes short reasoning first, then the answer

Models: **Gemini** (Colab built-in, no API key) and **qwen2.5:1.5b** running with **Ollama** on the Colab CPU. Total: 2 models x 3 styles x 50 tickets = 300 answers.

3. Results

Model	Prompt	Urgency %	Department %	Both %	Latency (s)	Input tokens
Gemini	Zero-shot	92	98	92	5.3	111
Gemini	Few-shot	92	100	92	5.2	277
Gemini	CoT	86	98	86	6.3	148
Qwen 1.5B	Zero-shot	60	72	42	0.2	130
Qwen 1.5B	Few-shot	68	66	38	0.2	295
Qwen 1.5B	CoT	48	66	28	0.6	165

Invalid JSON outputs: **0 of 300**. Retries needed: **0**. Cost: $0 (free tier and local model).

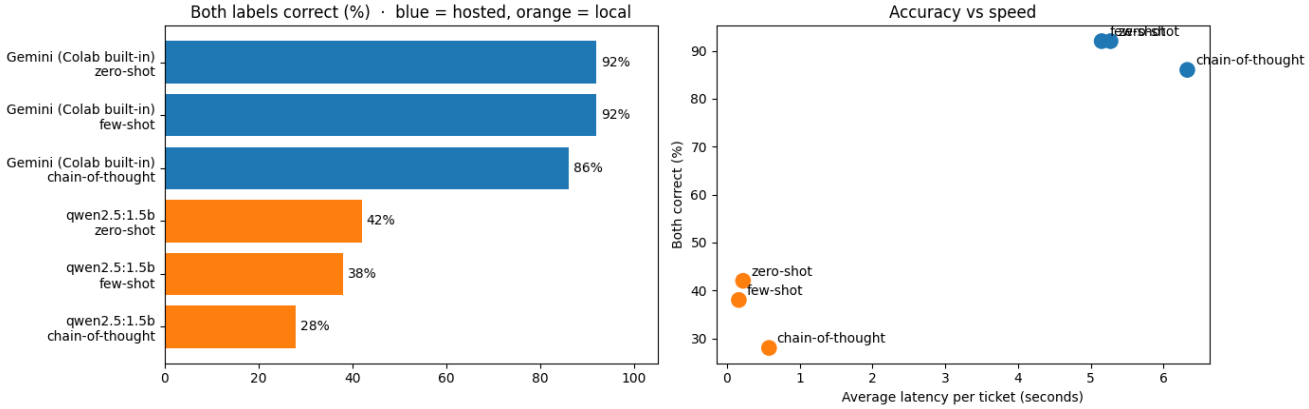


Figure 1. Accuracy (both labels correct) and accuracy vs speed

4. Key findings

- **Structured output works:** giving the model the JSON schema meant every answer was valid JSON, so the retry logic was never needed.
- **Gemini is far more accurate** (92% vs 42% at best). The small local model is fast and free but not reliable enough for this task.
- **Best accuracy for the cost: Gemini zero-shot**. Few-shot gave the same 92% but used 2.5x more input tokens; it did fix the one department mistake (100%).
- **Chain-of-thought did not help** here: lower accuracy, more output tokens (87 vs 11) and slower. The reasoning made the model over-rate urgency (for example, a late refund became "high").
- Most Gemini mistakes were **urgency, not department**, and the same tickets failed in every style (for example "Please delete my account and all my personal data": label medium, model low). Some of my own urgency labels are debatable, so urgency is partly subjective.

5. Limitations and next steps

- Only 50 tickets and one person labelled them; a second labeller would make the gold set fairer.
- Token counts for Colab's built-in Gemini are estimated (characters / 4).
- Next: use a Gemini API key for exact tokens and real prices, and try a bigger local model (for example 7B) if a GPU is available.